Uncertainty-Aware Frequency Constrained Scheduling with DNN Gradient Guidance Considering Wind Power Flexibility

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***Abstract*—With the increasing integration of wind power and the gradual retirement of conventional generators, maintaining frequency security has become a critical challenge in modern power systems. This paper develops an uncertainty-aware frequency constrained scheduling method with deep neural network gradient guidance (UFCS-DNN), which jointly optimizes the energy and reserve dispatch of conventional generation units (CGUs) and wind turbines (WTs). We embed the frequency nadir (FN) constraint by leveraging the gradient information of a trained DNN to iteratively integrate it into the scheduling model. In addition, a biased linear support vector machine (SVM) is employed to generate an initial solution close to the FN secure region, and a tailored strategy is designed to further improve the iterative convergence efficiency. To handle wind power uncertainty, a distributionally robust chance-constrained (DRCC) approach is adopted. Furthermore, the flexibility of WTs is fully exploited, including their capability to provide frequency support under configurable control modes and to act as regulation resources that dynamically respond to the uncertainty of other WTs. Case studies on the IEEE 5-bus and 118-bus test systems demonstrate that the proposed UFCS-DNN method successfully maintains the FN above 49.5 Hz across all time periods under predefined disturbances. Moreover, the acceleration strategy reduces the number of iterations by up to 90%. The flexible operation of WTs is effectively utilized to coordinate both frequency support and uncertainty response.**

***Index Terms*—Deep neural network, frequency constrained scheduling, frequency nadir constraint, distributionally robust chance constraint.**

NOMENCLATURE

Acronyms

CGU	Conventional generation unit.
PFR	Primary frequency response.
WT	Wind turbine.
FN	Frequency nadir.
RoCoF	Rate of change of frequency.
QSSFD	Quasi-steady-state frequency deviation.
SVM	Support vector machine.
MISOCP	Mixed-integer second-order cone programming.
DNN	Deep neural network.
UR	Uncertainty realization.
DRCC	Distributionally robust chance-constrained.

Variables

H_t^{sys}	System inertia at period t.
D_t^{sys}	System damping coefficient at period t.
$H_{w,t}$	Virtual inertia provided by WT w at period t.
$D_{w,t}$	Virtual damping provided by WT w at period t.
$R_{g,t}^G$	PFR reserve of CGU g at period t.
$R_{w,t}^W$	PFR reserve of WT w at period t.
$P_{g,t}$	Active power output of CGU g at period t.
$W_{w,t}$	Active power output of WT w at period t.
$\delta_{w,t}$	Deloading ratio of WT w at period t.
$\alpha_{g,w,t}$	Participation factor of CGU g at period t.
$\beta_{w,j,t}$	Participation factor of WT w at period t.
$R_{g,t}^+$, $R_{g,t}^-$	Upward and downward regulation reserves of CGU g at period t, respectively.
$\vartheta_{w,t}^{GFM}$, $\vartheta_{w,t}^{GFL}$	Binary indicator variables representing the GFM/GFL control modes of WT w at period t.
$u_{g,t}$	On/off status of CGU g at period t.
$z_{g,t}^u$, $z_{g,t}^d$	Start-up and shut-down status of CGU g at period t, respectively.

This work was supported by Smart Gird National Science and Technology Major Project (2030) under Grant 2025ZD0807200. *(Corresponding author: Zhigang Wu, e-mail: epzgwu@scut.edu.cn)*

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I. INTRODUCTION

A. Background and Motivation

THE global push for carbon emission reduction and climate change mitigation has significantly accelerated the penetration of wind power [1]. According to the National Energy Administration of China [2], as of the end of December 2024, the country's total installed wind power capacity reached approximately 520 GW, representing an 18.0% year-on-year increase. This rapid growth in renewable integration, coupled with the gradual retirement of conventional generation units (CGUs), has raised growing concerns over frequency security. In particular, the inherent uncertainty of wind power introduces considerable challenges to maintaining frequency stability and fully utilizing renewable generation.

To address these issues, frequency constrained scheduling (FCS) has been employed to co-optimize inertia and primary

frequency response (PFR) reserves, thereby enhancing the system's frequency response capability at the operational level [3]. However, the efficient incorporation of the frequency nadir (FN) constraint into optimization models remains a persistent challenge due to its nonlinearity. Meanwhile, wind turbines (WTs) can enhance system flexibility by providing reserves through deloading operation. While this helps reduce the reserve burden on CGUs, determining the optimal deloading coefficients becomes particularly challenging under high uncertainty.

Motivated by the above, this paper aims to address two key challenges in FCS with high wind penetration: (i) *How FN constraints can be efficiently embedded into the scheduling model*; and (ii) *How wind power uncertainty can be managed while effectively utilizing WT flexibility through deloading*.

B. Literature Review and Research Gaps

1) Efficient Embedding of the FN Constraint: The FN is obtained by solving a set of differential equations, which is inherently difficult to integrate into the convex optimization problem. By employing a linear approximation of the PFR process, the inertia-dependent PFR reserves constraint is used as a conservative approximation of the FN constraint [4]. Furthermore, by neglecting load damping and considering the combined effect of distinct PFR delivery services, a mixed-integer second-order cone (MISOCP) formulation of the FN constraint is derived [5]. The aforementioned studies construct FN constraints through simplifications and mathematical derivations, fundamentally following a model-driven method.

Recently, data-driven stability and security rules have been employed as surrogates for complex dynamic response processes in power systems [6]. In FCS studies, piecewise linearization (PWL) [7]–[9] has gained significant attention. By partitioning the dataset into multiple subsets and performing linear regression on each, PWL effectively approximates the FN constraint when the number of input features is relatively small. Support vector machines (SVMs) have also been widely used in FCS as linear surrogates for the nonconvex FN constraint [10], [11]. A convex hull approximation method has also been proposed to linearize the FN constraint while ensuring feasibility [12], [13]. *Although the data-driven models developed in these studies are widely adopted to extract FN constraints, their approximation accuracy deteriorates as the number of diversified frequency support resources increases, due to the growing dimensionality of the input space.*

Deep neural network (DNN) offers superior approximation capability, and its generalization can be enhanced by incorporating physical information [14]. Notably, ReLU-activation-based DNNs can be directly embedded into convex optimization models, making them well suited for mixed-integer linear surrogates of the nonconvex FN constraint. Currently, DNN-based FN constraints have been applied in various scenarios, including real-time FCS [15], [16], microgrid FCS [17], frequency constrained unit commitment [18], two-stage stochastic FCS [13], and nodal RoCoF-constrained scheduling [19]. However, the mixed-integer embedding method, which encodes ReLU activation functions with binary variables and formulates hidden layers as mixed-integer constraints, enables the use of commercial solvers but significantly complicates computation due to the increased problem complexity. Although some studies focus on pruning DNNs [20]–[22], they do not eliminate the need for embedding mixed-integer linear constraints, limiting scalability to larger systems. Recent study suggests that DNN-based stability constraints can be iteratively embedded in optimization problems in a linearized form by leveraging gradient computations [23]. This method begins from a conservatively stable operating point and gradually approaches the stability boundary, reducing operating cost while ensuring that stability constraints are not violated throughout the process. However, obtaining an initial operating point is far from trivial, and each iterative embedding of DNN gradient information requires solving a full scheduling problem. These factors necessitate faster convergence mechanisms for scalability to larger systems, and motivate the methodological developments of this work.

2) Wind Power Uncertainty Management and Flexibility Exploitation: The high uncertainty of wind power imposes greater demands on the regulation capability of power systems. In the day-ahead stage, a certain amount of regulation reserves is allocated to address intra-day uncertainty realization (UR), distinguishing it from PFR reserves, which are used to mitigate frequency decline under contingencies. Recently, various uncertainty optimization approaches have been adopted to manage uncertainty. Stochastic optimization based on multiple representative scenarios has been used to characterize wind power uncertainty [13]. However, as the number of scenarios increases, the problem size grows significantly, and the robustness of stochastic optimization requires further investigation. A two-stage chance-constrained stochastic optimization model has been proposed to address the uncertainty of wind-based virtual inertia [24]. However, the assumption that wind power forecasting errors follow a Gaussian distribution often fails to align with real-world conditions. Robust optimization has also been applied to mitigate the uncertainty of wind reserves [25], solved using the column-and-constraint generation algorithm. While this approach ensures system security, it tends to be overly conservative in practice and may even become infeasible for systems with high wind penetration.

In recent years, distributionally robust chance-constrained (DRCC) approaches have gained significant attention due to their ability to flexibly balance robustness and economic efficiency. Unlike conventional robust optimization, DRCC constructs uncertainty sets for probability distributions rather than direct forecasting errors, leveraging historical information such as mean and covariance. Various ambiguity set formulations have been used to characterize distributional uncertainty, including moment-based approaches [26], inexact moment-based approaches incorporating unimodality-skewness characteristics [27], Wasserstein metric-based approaches [28], Gaussian-centered Wasserstein-based approaches [29], and approaches incorporating heterogeneous uncertainty information [30]. The aforementioned studies primarily focus on the construction of ambiguity sets, while the adequacy of system regulation capability remains largely unaddressed. When wind power penetration is extremely high, fully responding

to UR may impose a significant economic burden and even render scheduling infeasible. In such cases, allowing WTs to operate in a deloaded mode can alleviate this issue and enhance flexibility by: (i) reducing the extent of UR response, (ii) enabling WTs to provide upward regulation reserves to compensate for the UR of other WTs, and (iii) contributing to PFR reserves. We observe that within the current DRCC research framework, Ref. [31] only considers (i) but neglects the provision of upward regulation reserves by WTs. Ref. [32] incorporates (ii) but assumes that UR must be fully addressed, while disregarding the frequency support capability of WTs. Ref. [33] solely accounts for virtual inertia and PFR reserves provided by WTs, with the deloading coefficient fixed at 0.1 rather than being optimized. *Overall, the joint optimization of WT deloading levels, upward regulation reserves, and PFR reserves within the DRCC framework has not been adequately explored, limiting the effective utilization of WT flexibility, and constitutes a key research gap addressed in this work.*

C. Contributions

Based on the above literature review and identified research gaps, we propose an uncertainty-aware frequency constrained scheduling with DNN gradient guidance (UFCS-DNN) method to address frequency security concerns, high uncertainty, and the exploitation of wind power flexibility. The main contributions are as follows:

1) Compared to existing methods that directly embed DNNs [13]–[22], we propose a method that iteratively incorporates linear constraints into FCS using DNN gradients as a surrogate for the nonconvex FN constraint. Additionally, in contrast to the latest research [23], we introduce acceleration techniques aimed at improving both the quality of the initial solution and the convergence.

2) Unlike existing approaches [31]–[33], which capture only partial aspects of WT flexibility, this work provides a refined formulation of WT deloaded power. This formulation enables WTs to simultaneously provide regulation reserves for balancing uncertainties arising from their own output as well as other WTs, and to allocate frequency response reserves to enhance system frequency security.

Fig. 1 illustrates the proposed UFCS-DNN framework, which consists of three main components. First, the input parameters of the scheduling system are collected, including the network topology, parameters of CGUs, forecasted wind and load power, and frequency limits. Based on these inputs, samples are generated within the parameter space of the studied system and used for DNN training.

Second, a DNN is trained to learn the nonlinear mapping between system operating conditions and the FN. After training, the scheduling problem without the complex FN constraint is solved to obtain candidate scheduling decisions, which are then fed into the trained DNN to evaluate the FN value and its corresponding gradients.

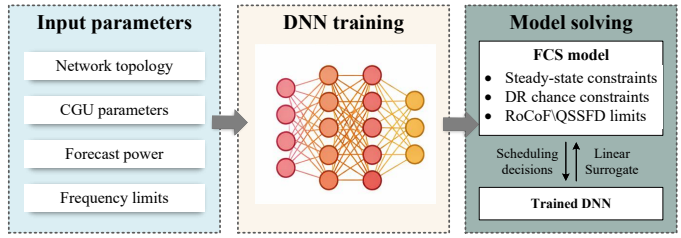

Fig. 1. Framework of the proposed UFCS-DNN.

Finally, the DNN gradients are exploited to construct linear surrogate FN constraints, guiding the subsequent optimization iterations toward the true FN security boundary. By iteratively integrating these linear surrogates into the scheduling model, the proposed framework progressively enforces the FN constraint without explicitly embedding the DNN into the optimization problem.

The remainder of this paper is organized as follows: Section II models the frequency dynamics; Section III presents the formulation of UFCS-DNN; the solution methodology is detailed in Section IV; Section V and VI present case studies on two test systems of different scales; and Section VII concludes the paper.

II. Modeling of Frequency Dynamics

A. Frequency Dynamics Description After Contingency

Consistent with the majority of FCS studies, we primarily focus on the dynamic process of the center-of-inertia frequency. After a contingency, the system frequency deviation can be described by the swing equation (1):

$$\frac{2H_t^{sys}}{f_0}\frac{\mathrm{d}\Delta f(\tau)}{\mathrm{d}\tau}+D_t^{sys}\Delta f(\tau)=\Delta P_t^G(\tau)+\Delta P_t^W(\tau)-\Delta P_t^{imb} \tag{1}$$

where t is the index of the scheduling period, H_t^{sys} represents the system inertia, f_0 is the nominal frequency, $\Delta f(\tau)$ is the frequency deviation, and D_t^{sys} denotes system damping (droop); ΔP_t^{imb} represents the power imbalance, $\Delta P_t^G(\tau)$ is the total PFR output of CGUs, and $\Delta P_t^W(\tau)$ is the total PFR output of WTs.

System operators typically do not rely on detailed PFR control parameters for scheduling but approximate the PFR process as a linear delivery [34], expressed as:

$$\begin{cases} \Delta P_t^G(\tau)=\sum\limits_{g\in\mathcal{G}}\min\left(\frac{R_{g,t}^G}{T_g}(\tau-\tau_g),R_{g,t}^G\right) \\ \Delta P_t^W(\tau)=\sum\limits_{w\in\mathcal{W}}\min\left(\frac{R_{w,t}^W}{T_w}(\tau-\tau_w),R_{w,t}^W\right) \end{cases} \tag{2}$$

where g and $\mathcal{G}$ denote the index and set of CGUs, respectively, while w and $\mathcal{W}$ represent the index and set of WTs, respectively. $R_{g,t}^G$, τ_g, and T_g correspond to the PFR reserve, dead time, and delivery time of CGUs, with analogous definitions for WTs, which are omitted for brevity. In this study, we assume: $\tau_w \le \tau_g \le \tau_w + T_w \le \tau_g + T_g, \quad \forall g, w$ to facilitate the derivation of frequency deviation expressions.

By combining (1) and (2), and arranging the PFR services in ascending order based on their dead time and delivery time

from the 1st to the Nth, we analytically derive the time-domain expression for frequency deviation:

$$\Delta f(\tau) = \begin{cases} \frac{\eta_1}{b}\left(1-e^{-(b/a)\tau}\right), & \tau < \tau_1 \\ \lambda_i e^{-(b/a)\tau} + \frac{\nu_i}{b}\tau + \frac{\eta_{i+1}b - a\nu_i}{b^2}, & \tau_i \le \tau < \tau_{i+1} \\ \lambda_{2N} e^{-(b/a)\tau} + \frac{\eta_{2N+1}}{b}, & \tau_{2N} \le \tau \end{cases} \tag{3}$$

where $i \in [1, 2N-1]$, and the detailed expressions of intermediate variables are provided in Appendix A.

B. Frequency Support Modeling of WTs

We consider that WTs can flexibly operate in grid-forming mode to provide virtual inertia and damping or in grid-following mode to deliver fast PFR power. This aligns with the practical operation of software-defined grids discussed in recent studies [35]. Based on this, H_t^{sys} and D_t^{sys} can be expressed as:

$$\begin{cases} H_t^{sys} = u_{g,t} H_g + \vartheta_{w,t}^{GFM} H_{w,t} \\ D_t^{sys} = D P_t^D + \vartheta_{w,t}^{GFM} D_{w,t} \end{cases} \tag{4}$$

where $u_{g,t}$ represents the online status of CGU, and H_g denotes its inertia. $\vartheta_{w,t}^{GFM} = 1$ indicates that the WT operates in grid-forming mode, $H_{w,t}$ and $D_{w,t}$ represent the virtual inertia and damping provided by WT, respectively. D and P_t^D denote the load damping coefficient and load demand, respectively.

Similarly, when WTs operate in grid-following mode, they are assumed to provide fast PFR services:

$$R_{w,t}^W \le \vartheta_{w,t}^{GFL} R_w^{\max} \tag{5}$$

where $\vartheta_{w,t}^{GFL} = 1$ indicates that the WT operates in grid-following mode, and $R_w^{\max}$ represents the upper limit of its fast PFR reserves.

The control mode of WTs can only be in one state within a given period, which is expressed as:

$$\vartheta_{w,t}^{GFL} + \vartheta_{w,t}^{GFM} = 1, \quad \forall w, t \tag{6}$$

C. Modeling of Frequency Constraints

In FCS studies, the commonly considered frequency constraints include the initial rate of change of frequency (RoCoF), FN, and quasi-steady-state frequency deviation (QSSFD). Among these, the RoCoF and QSSFD constraints can be readily formulated in a linear form:

$$\frac{\Delta P_t^{imb} f_0}{2H_t^{sys}} \le \overline{RoCoF}, \forall t \tag{7}$$

$$\Delta P_t^{imb} \le D_t^{sys} \Delta f_{\max}^{qss} + \sum_{g\in\mathcal{G}} R_{g,t}^G + \sum_{w\in\mathcal{W}} R_{w,t}^W, \forall t \tag{8}$$

where $\overline{RoCoF}$ and $\Delta f_{\max}^{qss}$ are the limits of RoCoF and QSSFD, respectively.

The extremum of $\Delta f(\tau)$ requires piecewise analytical evaluation, making the FN constraint challenging to handle. We first express it in a compact form as:

$$\chi\left(H_t^{sys} D_t^{sys} \mathbf{R}_t^G, \mathbf{R}_t^W\right) \le \Delta f_{\max}, \forall t \tag{9}$$

where $\Delta f_{\max}$ represents the FN limit, $\mathbf{R}_t^G$ and $\mathbf{R}_t^W$ denote the reserve capacity vectors of CGUs and WTs at time t, respectively, where $\mathbf{R}_t^G = \{R_{g,t}^G\}_{g\in\mathcal{G}}$ and $\mathbf{R}_t^W = \{R_{w,t}^W\}_{w\in\mathcal{W}}$. The detailed derivation of the FN constraint is presented in Appendix A.

III. Formulation of UFCS-DNN

A. Uncertainty Modeling

1) Wind Power Forecast Uncertainty: The actual wind power output $\tilde{W}_{w,t}$ consists of the day-ahead forecasted value $W_{w,t}^f$ and the forecast error $\varpi_{w,t}$. Considering that WTs can operate in the deloaded mode to enhance flexibility, $\tilde{W}_{w,t}$ is modeled as:

$$\tilde{W}_{w,t} = (1-\delta_{w,t})\left(W_{w,t}^f - \varpi_{w,t}\right), \quad \forall w, t \tag{10}$$

where $\delta_{w,t}$ represents the deloading coefficient of WTs, constrained within the range $[0, 1]$.

2) Refined Reserve Partitioning for WTs: The deloaded power is divided into two components: upward regulation reserves and PFR reserves:

$$\begin{cases} \tilde{R}_{w,t}^+ = \delta_{w,t}^+ \left(W_{w,t}^f - \varpi_{w,t}\right) \\ \tilde{R}_{w,t}^{PFR} = \delta_{w,t}^{PFR} \left(W_{w,t}^f - \varpi_{w,t}\right) \end{cases} \tag{11}$$

where $\tilde{R}_{w,t}^+$ represents the upward regulation reserve provided by WTs, and $\tilde{R}_{w,t}^{PFR}$ denotes the PFR reserve. The sum of their corresponding reserve coefficients $\delta_{w,t}^+$ and $\delta_{w,t}^{PFR}$ is constrained by the total deloading coefficient $\delta_{w,t}$:

$$\delta_{w,t}^+ + \delta_{w,t}^{PFR} \le \delta_{w,t}, \forall w, t \tag{12}$$

3) Modeling of Uncertainty Realization Response: CGUs can adjust their output following the affine policy to respond to UR:

$$\tilde{P}_{g,t} = P_{g,t} + \sum_{w\in\mathcal{W}} \alpha_{g,w,t} \varpi_{w,t}, \forall g, t \tag{13}$$

where $\tilde{P}_{g,t}$ represents the actual power output after UR, $P_{g,t}$ denotes the scheduled power output, and $\alpha_{g,w,t}$ is the participation factor of CGU g in responding to the UR of WT w.

In this paper, we consider that WTs respond to UR through the upward regulation reserve from their deloaded power. Specifically, the actual power output of WTs is reformulated as:

$$\tilde{W}_{w,t} = (1-\delta_{w,t})\left(W_{w,t}^f - \varpi_{w,t}\right) + \sum_{w\neq j} \beta_{w,j,t} \varpi_{j,t}, \forall w, t \tag{14}$$

where $\beta_{w,j,t}$ is the participation factor of WT w in responding to the UR of WT j.

B. Objective Function

UFCS-DNN aims to minimize the operational cost, which includes the no-load, startup, shutdown, generation, upward

regulation reserve, and PFR reserve costs of CGUs, as well as the deloading cost of WTs, as shown in (15):

$$\begin{aligned}&\min \sum_{t\in\mathcal{T}}\sum_{g\in\mathcal{G}}\left(c_g^{NL}u_{g,t}+c_g^{SU}z_{g,t}^{u}+c_g^{SD}z_{g,t}^{d}+c_g^{GEN}P_{g,t}\right)+\\&\sum_{t\in\mathcal{T}}\sum_{g\in\mathcal{G}}\left(c_g^{+}R_{g,t}^{+}+c_g^{-}R_{g,t}^{-}+c_g^{PFR}R_{g,t}^{G}\right)+\\&\sum_{t\in\mathcal{T}}\sum_{w\in\mathcal{W}}c_w^{De}\delta_{w,t}W_{w,t}^{f}\end{aligned} \tag{15}$$

where c_g^{NL}, c_g^{SU}, c_g^{SD}, c_g^{GEN}, c_g^{+}, c_g^{-}, and c_g^{PFR} represent the no-load, startup, shutdown, generation, upward regulation reserve, downward regulation reserve, and PFR reserve prices of CGU g, respectively. $z_{g,t}^{u}$ and $z_{g,t}^{d}$ denote the startup and shutdown status of CGU g, respectively. $R_{g,t}^{+}$ and $R_{g,t}^{-}$ correspond to the upward and downward regulation reserves of CGU g. c_w^{De} represents the deloading price of WTs.

C. Constraints

1) CGUs Constraints: The constraints of CGUs include startup and shutdown logic constraints, power limits, ramping constraints, and minimum up/down time constraints. The detailed mathematical formulations can be referred to in [7].

Here, we focus on the upward and downward regulation reserve constraints of CGUs, which are modeled as DR chance constraints:

$$\inf_{\mathbb{P}\in\mathcal{P}_t}\left\{\alpha_{g,t}^{\mathrm{T}}\varpi_t\le R_{g,t}^{+}\right\}\ge 1-\varepsilon_g,\forall g,t \tag{16}$$

$$\inf_{\mathbb{P}\in\mathcal{P}_t}\left\{-\alpha_{g,t}^{\mathrm{T}}\varpi_t\le R_{g,t}^{-}\right\}\ge 1-\varepsilon_g,\forall g,t \tag{17}$$

where $\alpha_{g,t}=\left[\begin{array}{ccc}\alpha_{g,1,t} & \cdots & \alpha_{g,|\mathcal{W}|,t}\end{array}\right]^{\mathrm{T}}$ and $\varpi_t=\left[\begin{array}{ccc}\varpi_{1,t} & \cdots & \varpi_{|\mathcal{W}|,t}\end{array}\right]^{\mathrm{T}}$. These constraints ensure regulation reserve adequacy with a confidence level of at least $1-\varepsilon_g$ under the worst-case probability distribution within the ambiguity set $\mathcal{P}_t$.

2) WTs Constraints: The operational constraints of WTs include the fast PFR constraint in GFL control mode (5), the control mode constraint (6), the deloading constraint (12), as well as the regulation reserve constraints (18) and the PFR reserve constraint (19), which are modeled as DR chance constraints:

$$\inf_{\mathbb{P}\in\mathcal{P}_t}\left\{\beta_{w,t}^{\mathrm{T}}\varpi_t\le\tilde{R}_{w,t}^{+}\right\}\ge 1-\varepsilon_w,\forall w,t \tag{18}$$

$$\inf_{\mathbb{P}\in\mathcal{P}_t}\left\{\vartheta_{w,t}^{GFM}P_{w,t}^{GFM}+\vartheta_{w,t}^{GFL}R_{w,t}^{W}\le\tilde{R}_{w,t}^{PFR}\right\}\ge 1-\varepsilon_w,\forall w,t \tag{19}$$

where $P_{w,t}^{GFM}=2H_{w,t}\overline{RoCoF}+D_{w,t}\Delta f_{\max}$ represents a conservative approximation of the reserve required for WTs operating in the GFM control mode to provide virtual inertia and damping support [12], $\beta_{w,t}=\left[\begin{array}{ccc}\beta_{w,1,t} & \cdots & \beta_{w,|\mathcal{W}|,t}\end{array}\right]^{\mathrm{T}}$ and $\beta_{w,w,t}=0,\quad\forall w,t$.

3) System Constraints: The system constraints include the power balance constraint after UR and the AC line capacity constraint, as shown in (20) and (21)-(22), respectively:

$$\sum_{g\in\mathcal{G}}\tilde{P}_{g,t}+\sum_{w\in\mathcal{W}}\tilde{W}_{w,t}=\sum_{d\in\mathcal{D}}P_{d,t}^{D},\forall t \tag{20}$$

$$\inf_{\mathbb{P}\in\mathcal{P}_t}\left\{\tilde{f}_{l,t}\le f_l^{\max}\right\}\ge 1-\varepsilon_l,\forall l,t \tag{21}$$

$$\inf_{\mathbb{P}\in\mathcal{P}_t}\left\{-f_l^{\max}\le\tilde{f}_{l,t}\right\}\ge 1-\varepsilon_l,\forall l,t \tag{22}$$

where $\tilde{f}_{l,t}=\sum_{g\in\mathcal{G}}M_{g,l}\tilde{P}_{g,t}+\sum_{w\in\mathcal{W}}M_{w,l}\tilde{W}_{w,t}-\sum_{d\in\mathcal{D}}M_{d,l}P_{d,t}^{D}$, d and $\mathcal{D}$ represent the index and set of loads, respectively, and l denotes the index of transmission lines. $P_{d,t}^{D}$ represents the load power, and $M_{(\cdot),l}$ denotes the power transfer distribution factor (PTDF).

By separating the random variables in (20), we derive its corresponding equivalent deterministic constraint, the derivation process is provided in the Appendix B.

$$\sum_{g\in\mathcal{G}}P_{g,t}+\sum_{w\in\mathcal{W}}W_{w,t}=\sum_{d\in\mathcal{D}}P_{d,t}^{D},\forall t \tag{23}$$

$$\sum_{g\in\mathcal{G}}\alpha_{g,w,t}+\sum_{j\in\mathcal{W}}\beta_{j,w,t}=1-\delta_{w,t},\forall w,t \tag{24}$$

Remark1: As observed from (24), the flexibility of WTs in responding to UR is reflected in two aspects: reducing their own UR response through deloading coefficients and reserving (upward) regulation reserves to respond to the UR of other WTs. This dual effect significantly alleviates the burden on CGUs in handling UR. Furthermore, since WTs typically do not operate far from their maximum power point, which can be controlled by limiting the maximum deloading coefficient, we assume they possess sufficient downward regulation reserves.

4) Frequency Constraints: The frequency constraints include RoCoF (7), QSSFD (8), and FN (9), as presented in Section II.

IV. Solution Methodology

The UFCS-DNN established in Section III cannot be directly solved by commercial solvers due to the presence of (i) DR chance constraints and (ii) FN constraints. This section introduces the methodologies for handling these challenges.

A. Linear Reformulation of DR Chance Constraints

To address the DR chance constraints, we adopt a Wasserstein-based ambiguity set to characterize the uncertainty in wind power forecast errors. By leveraging conditional value-at-risk (CVaR) approximation and duality theory, a DR chance constraint of the form $\inf_{\mathbb{P}\in\mathcal{P}}\left\{a^{\mathrm{T}}(x)\xi+b(x)\le 0\right\}\ge 1-\varepsilon$ can be reformulated into the following linear constraints [28]:

$$\left\{\begin{array}{l}\lambda\theta-\beta\varepsilon+\frac{1}{|\mathcal{N}|}\sum_{n\in\mathcal{N}}z_n\le 0\\ z_n\ge a^{\mathrm{T}}(x)\hat{\xi}_n+b(x)+\beta,\forall n\\ \|a(x)\|_{\infty}\le\lambda\\ \lambda,\beta,z_n\ge 0,\forall n\end{array}\right. \tag{25}$$

For more detailed information on (25), please refer to [28].

B. DNN Gradient Guided Linear Surrogate of FN Constraints

Next, we introduce an iterative algorithm guided by DNN gradients to progressively enforce the FN constraint. The DNN is trained using the mean squared error (MSE) as the loss function and employs ReLU activation functions to approximate the FN value, denoted by $\chi(\mathbf{x})$. The approximated

function is denoted as $\hat{\chi}(\mathbf{x})$. The DNN gradient is computed via backpropagation, obtaining the partial derivatives of the output with respect to the input features. We then use this gradient to construct a linear surrogate, iteratively updating the approximation of the nonconvex FN constraint [23]:

$$\hat{\chi}\left(\mathbf{x}_k^*\right)+\nabla\hat{\chi}\left(\mathbf{x}_k^*\right)\left(\mathbf{x}_{k+1}-\mathbf{x}_k^*\right)\leq\Delta f_{\max} \tag{26}$$

where $\mathbf{x}_k^*$ is the solution at the kth iteration, $\nabla\hat{\chi}\left(\mathbf{x}_k^*\right)$ represents the gradient at the kth iteration, $\mathbf{x}_{k+1}$ denotes the optimization variable for the $(k+1)$th iteration.

This approach effectively reduces the computational complexity while ensuring the FN constraint is progressively satisfied. However, the following issues still need to be addressed: (i) How should the initial solution be selected for the first iteration? A good initial point should be close to the security boundary. This not only facilitates a faster approach to the boundary but also reduces inaccuracies when the initial solution is far from the learned data [20]. (ii) How can convergence be ensured? We observe that the right-hand side of (26) is fixed at $\Delta f_{\max}$. As the solution methods this value, oscillations may occur. This paper proposes solutions to both of these open questions.

For (i), we propose a biased SVM-based approach to improve the quality of the initial solution in the first iteration. Specifically, we classify points that violate the FN constraint as label 1 and those that satisfy it as label 0. By adjusting the SVM to focus more on correctly classifying label 1, i.e., increasing precision, we obtain a relatively relaxed initial solution. Based on the above discussion, the linear constraints derived from the biased SVM are incorporated into the UFCS-DNN:

$$\omega^{\mathrm{T}}\mathbf{x}+b\leq 0 \tag{27}$$

where ω represents the slope, and b denotes the intercept.

For (ii), to suppress oscillations and ensure convergence, we modify the right-hand side coefficient of (26) in the later stages of the iteration:

$$\hat{\chi}\left(\mathbf{x}_k^*\right)+\nabla\hat{\chi}\left(\mathbf{x}_k^*\right)\left(\mathbf{x}_{k+1}-\mathbf{x}_k^*\right)\leq\left(1-\varsigma_{k+1}\right)\hat{\chi}(\mathbf{x}_k^*) \tag{28}$$

where ς_{k+1} is the step size, which controls the progress made towards the feasible region in each iteration. It can be computed as follows:

$$\varsigma_{k+1}=\min\left(\varsigma^{\max},\max\left(\begin{array}{l}\varsigma^{\min},\varsigma^{\min}+\left(\varsigma^{\max}-\varsigma^{\min}\right)\times\\ \left(1-\min\left(\frac{\eta_k}{\upsilon},1\right)\right)\end{array}\right)\right) \tag{29}$$

where $\eta_k=\left(\chi\left(\mathbf{x}_k^*\right)-\Delta f_{\max}\right)/\Delta f_{\max}$, υ is a hyperparameter, $\varsigma^{\max}$ and $\varsigma^{\min}$ are the upper and lower bounds of ς_{k+1}.

Finally, the iterative solution algorithm for UFCS-DNN is presented in Algorithm 1. We first solve the scheduling model with the SVM constraint set ϕ and check the satisfaction of FN constraints for each time period. If the computed $\chi\left(\mathbf{x}_{k,t}^*\right)$ is far from $\Delta f_{\max}$, we use Δf_{bd} as the threshold or consider whether the current iteration count has not reached the oscillation limit k_{lim}. In this case, we first incorporate (26) as a linear surrogate for the FN constraint. Otherwise, if the current solution $\mathbf{x}_{k,t}^*$ is deemed sufficiently close to the security boundary, we remove the SVM constraint and iteratively add (28) as the linear surrogate. This process continues until the conditions are met. Additionally, we recommend that the previously added (26) constraints be removed before the current optimization.

Remark 2: It is worth noting that, for the purpose of initialization, the solution only needs to be reasonably close to the FN security boundary, while the accuracy is progressively refined through the subsequent DNN-gradient-based iterations. Therefore, a linear SVM is adopted to enhance the quality of the initial solution due to its structural simplicity. Similarly, the piecewise linearization method proposed in [8] can also be employed to preliminarily learn the FN constraint. More sophisticated methods, such as our previous work using the SVM-based decision tree model [36], can achieve higher modeling accuracy; however, they inevitably lead to mixed-integer constraint structures, which are unfavorable for iterative solution procedures.

C. Discussion

1) Gradient Misguidance: We acknowledge that gradient inaccuracy is an inherent challenge when learning and approximating a nonsmooth FN boundary. Before applying any remedial actions, it is first necessary to identify whether the gradient guidance is indeed misleading. This can be achieved by monitoring the evolution of the FN value at each scheduling period across iterations. If, over multiple successive iterations, the FN values of certain periods consistently move away from the security boundary or exhibit persistent oscillations, this behavior can be interpreted as an indication of inaccurate or misleading gradient information. In such cases, two heuristic remedial strategies can be adopted. First, for the affected periods, the biased SVM initialization can be made more conservative to generate an operating point closer to the FN boundary, thereby reducing the reliance on inaccurate gradient guidance. Second, gradient-based guidance for these periods can be temporarily suspended and replaced by alternative local refinement strategies, such as quasi-Newton-type updates, to compensate for gradient misdirection.

We note that systematically detecting and correcting gradient misguidance in nonsmooth regions remains a challenging problem. The development of more effective strategies to mitigate inaccurate DNN gradient guidance is therefore left as an important direction for future research.

2) Recovery from SVM Misclassification: The role of the SVM in the proposed framework is to provide an initial operating point that is close to the FN security boundary, preferably slightly infeasible, so that the subsequent DNN-gradient-based iterations can efficiently refine the solution. When the SVM misclassifies the feasible region, it typically stems from an overly conservative separating hyperplane, which may place the initial solution well inside the FN-feasible region for some time periods. In such cases, the corresponding FN constraints are already satisfied, possibly with large safety margins, leading to conservative but robust initial decisions.

Importantly, the proposed iterative framework can naturally recover from this situation. If a more conservative operation is desired, the system operator may simply fix the decisions in those periods that already satisfy the FN constraint and

focus the iterations on the remaining infeasible periods. Alternatively, if a less conservative and more accurate solution is preferred, the SVM can be retrained by adjusting its bias. The iterative procedure can then be restarted using the updated SVM initialization. In both cases, the framework remains stable and flexible, ensuring reliable convergence without compromising feasibility.

3) Guarantee of FN Constraint Satisfaction: As shown in Algorithm 1, after each optimization step, the FN values of all time periods are validated using the accurate nonlinear FN constraint (9) (Step 9). This implies that the iterative process is terminated if and only if the accurate FN constraint is satisfied in all scheduling periods. Therefore, the final set of FN cuts incorporated in the scheduling model does not relax the true FN constraint. Furthermore, if enhanced robustness is desired, the right-hand sides of (26) or (28) can be appropriately tightened.

4) Computational Complexity of a Single Outer Iteration: The computational complexity of a single outer iteration is only weakly affected by the depth or width of the DNN. This is because the DNN is not embedded into the scheduling model; it is only used to evaluate gradients for constructing surrogate linear cuts in (26) or (28). With respect to surrogate constraints, at most T linear FN cuts are added in each iteration, where T denotes the number of scheduling periods. The inclusion of several tens of linear constraints introduces negligible overhead to the mixed-integer scheduling problem, even for large-scale instances. This observation is further supported by the numerical results presented in the case studies, where no significant increase in solution time is observed as the number of cuts grows.

Overall, the dominant factor affecting the computational efficiency of each iteration remains the size of the original scheduling model itself, such as the number of generating units, network scale, and associated operational constraints, rather than the DNN structure or the progressive addition of surrogate cuts.

Algorithm 1 Iterative Solution Algorithm for UFCS-DNN

Input: System parameters, DNN model, SVM constraints set ϕ, $\varsigma^{\max}$, $\varsigma^{\min}$

Output: Scheduling results satisfying FN constraint or infeasibility

1 **Initialization:** Iteration count $k \leftarrow 0$, convergence flag Flag $\leftarrow 1$, DNN constraints set $\mathcal{C} \leftarrow \emptyset$, and initialize the scheduling model Model.

2 **while *True* do**
3 Flag $\leftarrow 1$
4 Solve Model
5 Model.$remove(\mathcal{C})$, $\mathcal{C} \leftarrow \emptyset$
6 **if** *Solution is infeasible* **then**
7 **Exit**
8 **for** $t \in \mathcal{T}$ **do**
9 **if** $\chi\left(\mathbf{x}^*_{k,t}\right) \leq \Delta f_{\max}$ **then**
10 **Pass**
11 **else**
12 Flag $\leftarrow 0$
13 **if** $\chi\left(\mathbf{x}^*_{k,t}\right) \leq \Delta f_{bd}$ ***or*** $k \geq k_{\lim}$ **then**
14 Model.$remove(\phi_t)$, Model.$addConstr$(28);
15 **else**
16 $\mathcal{C} \leftarrow \mathcal{C} \cup \{(26)\}$ Model.$addConstr$(26)
17 **if** *Flag* $= 0$ **then**
18 $k \leftarrow k + 1$ **Restart iteration**
19 **else**
20 Save results and **Exit**

V. Case Study I: Modified IEEE 5-bus Test System

A. Test System Introduction and Parameter Settings

1) Modified IEEE 5-bus Test System Introduction : In this section, we perform case studies on the modified IEEE 5-bus test system. The topology of the modified IEEE 5-bus test system is shown in Fig. 2, where the installed capacity of CGUs is 720 MW, wind power capacity is 300 MW, and the load level ranges from 300 MW to 640 MW. More detailed information can be found in [37]. In this test system, all WTs are assumed to have identical deadband and delivery times of 0.1s and 2s, respectively, while CGUs have values of 1s and 10s.

2) DNN Parameters Setting: The feature dimension of the DNN is four, namely the system inertia, system damping, PFR reserve of CGUs, and PFR reserve of WTs. A total of 20,000 samples are generated using Latin Hypercube Sampling (LHS) within the lower and upper bounds of these four variables. The corresponding sampling ranges are summarized in Table I, where all variables are expressed in per-unit (p.u.) values with respect to the hourly load demand. The biased SVM is trained using the SVC model from `sklearn`, with a bias factor of 3.5 for label 0 and 1 for label 1. The DNN is trained using `PyTorch` with `epoch` set to 200, a learning rate of 0.0001, and a network architecture of `DNNlayer_sizes` = [4, 256, 128, 64, 32, 1], with a batch size of 32.

TABLE I
Sampling Ranges of DNN Input Features in the Modified IEEE 5-Bus System

Variable	H^{sys}	D^{sys}	R^W	R^G
Range	5–15 s	0.03–0.20 p.u.	0–0.08 p.u.	0–0.20 p.u.

3) Scheduling Model Parameters Setting: The scheduling period is 24 hours with a step size of 1 hour, we assume a power imbalance of 10% of the current period's load in each time period. The nominal frequency is 50 Hz, the limits of RoCoF, FN, and QSSFD are set to 1 Hz/s, 49.5 Hz (i.e., $\Delta f_{\max} = 0.5$ Hz), and 0.25 Hz, respectively. The Wasserstein ball radius is set to 0.5, and all DR chance constraints are required to be satisfied with a confidence level of 0.95, i.e., ε = 0.05 [33]. The oscillation limit $k_{\lim}$ is 10, $\varsigma^{\max}$ is 0.05, Δf_{bd} is 0.6 Hz, $\varsigma^{\min}$ is 0.01, and hyperparameter υ is 1.

Using the Monte Carlo method, 2,000 prediction error scenarios are generated for each WT, with a mean of 0

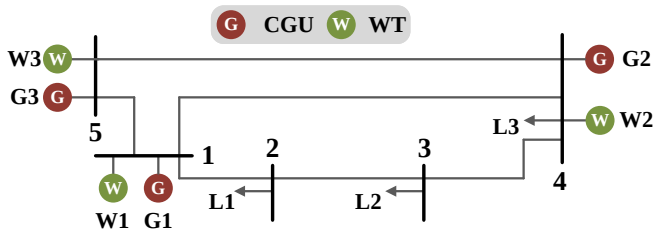

Fig. 2. Modified IEEE 5-bus test system.

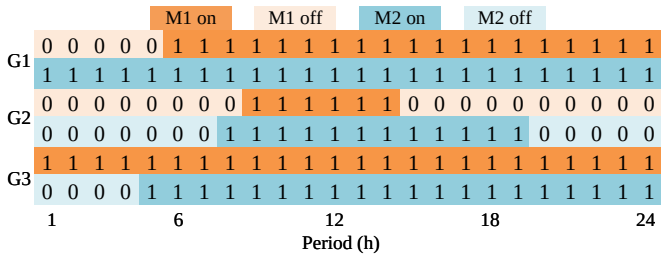

Fig. 3. CGUs commitment results of M1 and M2 in the modified IEEE 5-bus test system.

and a standard deviation of 5%. Among these, 20 scenarios are utilized to construct the distributionally robust chance constraint, while the remaining 1,980 scenarios are used to evaluate the confidence level of the chance constraint. The solving of UFCS-DNN is conducted on a computer with a 3.00 GHz Intel Core i5-8500 CPU and 16 GB of RAM, using Gurobi 11.0.0 as the solver and Python 3.10.6 as the programming language. The MIPGAP is set to 0.1%.

TABLE II
COST COMPARISON OF DIFFERENT METHODS IN THE MODIFIED IEEE 5-BUS TEST SYSTEM.

Cost (k$)	M1	M2
No load	23.12	27.20
Start-up/Shut-down	8.40	8.40
Generation	204.25	232.15
CGUs Regulation reserve	4.76	4.81
CGUs PFR reserve	28.40	8.51
WTs deloading	5.27	13.69
Total	274.20	294.76

B. Comparison with and without FN Constraint

We define two methods: M1 and M2, where M1 represents the UC method without FN constraints, and M2 corresponds to the proposed UFCS-DNN method, to analyze the impact of considering FN constraints.

1) Economic Cost Comparison: The economic costs under M1 and M2 are presented in Table II. It can be observed that due to the consideration of FN constraints, the total cost of M2 increases by 7.5% compared to M1. This is because M2 imposes stricter requirements on (i) system inertia and (ii) PFR reserves.

Specifically, in aspect (i), as shown in Fig. 3, the commitment results of CGUs indicate that G1, which has the highest inertia (280 MW × 9 s), is fully committed throughout the 24-hour period in M2. Additionally, G2, which has the second-highest inertia (240 MW × 8 s), is committed 6 hours longer in M2 than in M1, whereas G3, which has the lowest inertia (200 MW × 6 s), is committed less frequently in M2. Overall, the system inertia in M2 is consistently higher than in M1 across all time periods, as illustrated in Fig. 4. Consequently, the no-load and generation costs in M2 increase by 17.65% and 13.66%, respectively, compared to M1.

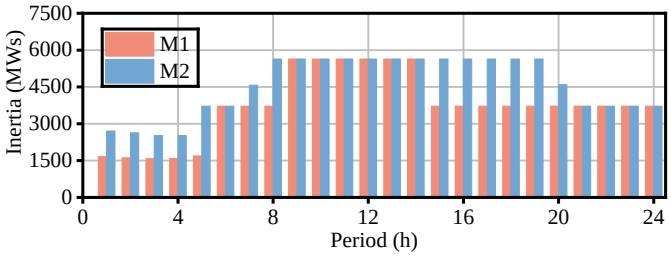

Fig. 4. System inertia of M1 and M2 in the modified IEEE 5-bus test system.

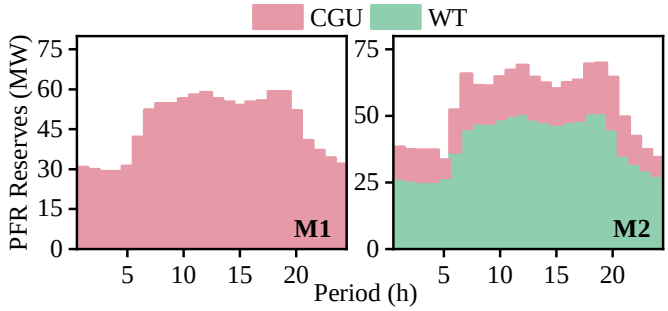

Fig. 5. PFR reserves deployment of M1 and M2 in the modified IEEE 5-bus test system.

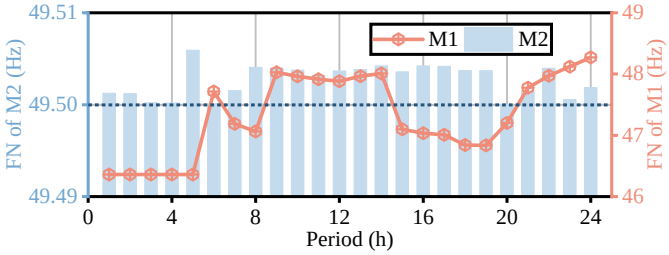

Fig. 6. FN under preset power imbalance for M1 and M2 in the modified IEEE 5-bus test system.

A more significant difference is observed in aspect (ii), i.e., PFR reserves. The PFR compositions under M1 and M2 are depicted in Fig. 5. With FN constraints considered, the PFR reserve provision from CGUs decreases by 70.04%, while the deloading cost of WTs increases by 159.78%. This is because the rapid PFR delivery capability of WTs makes them more effective in mitigating frequency drops for the same reserve capacity. In low-inertia conditions, the demand for fast frequency response resources becomes particularly critical. To ensure FN constraints are satisfied, a larger proportion of WTs' PFR reserves are scheduled, while slower-responding CGUs are less favored.

2) FN Comparison: We compute FN by substituting the results of M1 and M2 at each time period into $\chi(\mathbf{x})$, and the

comparison results are shown in Fig. 6. The blue dashed line, aligned with the left axis, represents the FN limit of 49.5 Hz. It can be observed that under the M1 method, FN violates the constraint in all time periods, ranging between 46 Hz and 49 Hz. It is important to note that $\chi(\mathbf{x})$ represents theoretical calculations and does not account for under frequency load shedding actions. In contrast, under the M2 method, FN is strictly maintained above 49.5 Hz. Moreover, FN does not exceed 49.5 Hz significantly, remaining below 49.51 Hz, indicating that the iteratively added linear surrogate is accurate and does not lead to conservative results.

C. Verification of the Effectiveness of the Accelerated Iterative Method

In this subsection, ablation studies are conducted to evaluate the contribution of different algorithmic components. Specifically, four variants are considered as benchmarks: 1) removing the SVM-based initial solution, 2) adopting an unbiased SVM as the initial solution, 3) using a fixed step size ($\varsigma^{\min}$) instead of the tailored step-size strategy, and 4) removing both the SVM-based initialization and the tailored step-size strategy. Fig. 7 illustrates the convergence trajectories of the total cost for these variants in comparison with the proposed method. The corresponding iteration numbers, initial objective values, and final total costs are summarized in Table III.

The ablation study results demonstrate the effectiveness of each component in the proposed framework. The proposed method converges within 15 iterations and achieves a low final total cost, indicating a good balance between convergence speed and solution quality. Removing the SVM-based initialization significantly increases the number of iterations (from 15 to 41), while further removing the tailored step-size leads to a dramatic deterioration in convergence efficiency, with up to 337 iterations required. This highlights the critical role of the SVM-based initial solution and the tailored step-size strategy in accelerating convergence.

In addition, adopting an unbiased SVM yields fast convergence but results in a noticeably higher final cost, because the resulting initial solution satisfies the FN constraint in some periods with excessive margins, i.e., it lies beyond the FN security boundary. As a consequence, unnecessary conservativeness is introduced at the initialization stage, leading to degraded economic performance. Overall, the ablation results confirm that both the biased-SVM-based initialization and the tailored step-size mechanism are essential for achieving fast and stable convergence with satisfactory optimality.

TABLE III
ABLATION STUDY RESULTS

Method	Iterations	Initial (k$)	Final (k$)
Proposed method	15	290.42	294.76
w/o SVM constraint	41	260.32	295.10
with unbiased SVM constraint	10	296.51	297.10
w/o tailored step-size	28	290.42	294.74
w/o SVM constraint and tailored step-size	337	260.32	294.71

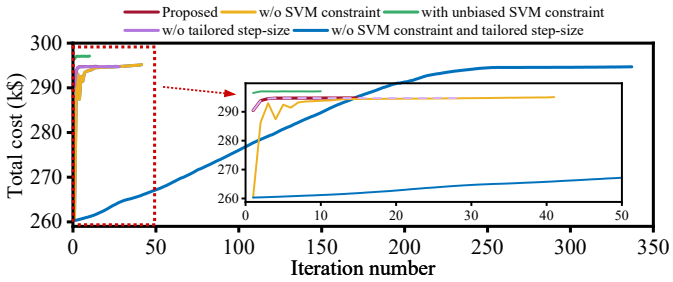

Fig. 7. Iterative process under different iterative methods.

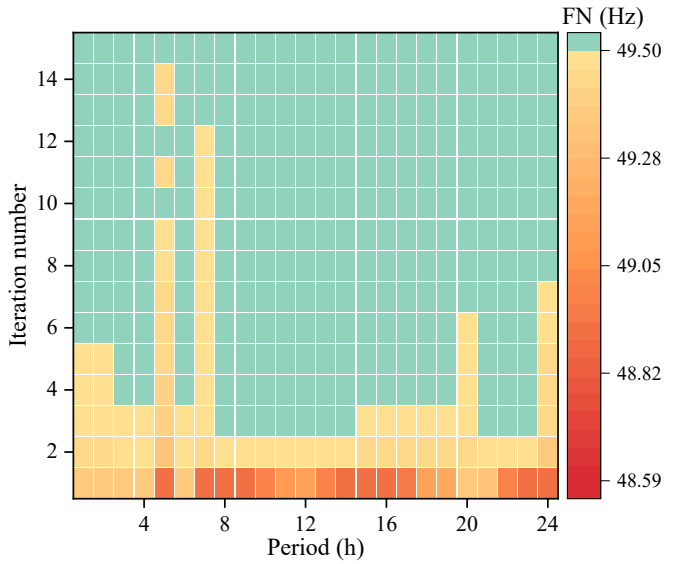

Fig. 8. FN across time periods during the iterative process in the modified IEEE 5-Bus test system.

D. Illustration of Iterative Details of the Proposed Method

Fig. 8 illustrates the FN values across different time periods during the iterative process in the modified IEEE 5-bus test system. It can be observed that, in the first iteration, the FN values in all time periods significantly exceed the security threshold of 49.5 Hz, which results in the introduction of 24 linear FN cuts, one for each period. After incorporating the DNN-gradient-based linear FN constraints, the FN violations are noticeably alleviated in the second iteration. In the third iteration, the FN constraint is already satisfied in several periods. As the iterations proceed, only a few periods remain infeasible, and from the eighth iteration onward, FN violations occur in only a small number of time periods. This behavior highlights the fast convergence property of the proposed method.

Fig. 9 further depicts the computational time of each iteration. It can be seen that no clear monotonic relationship exists between the solution time and the iteration index, indicating that the progressive addition of FN cuts does not lead to a significant increase in model size or computational burden.

E. Verification of WTs' Operational Flexibility

In this part, we investigate the impact of considering WTs providing upward regulation reserves on costs under different levels of wind power uncertainty, which is characterized by the

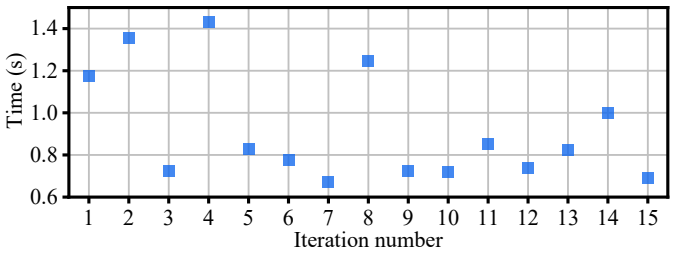

Fig. 9. Optimization solution time per iteration.

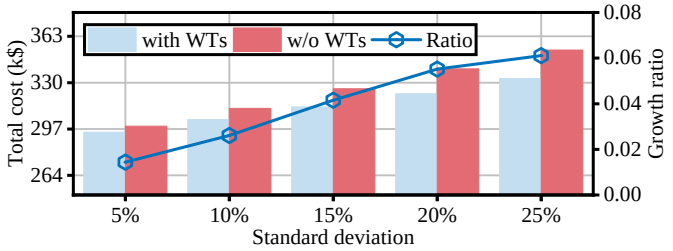

Fig. 10. Comparison of total cost with and without WTs providing upward regulation reserves under different standard deviation settings.

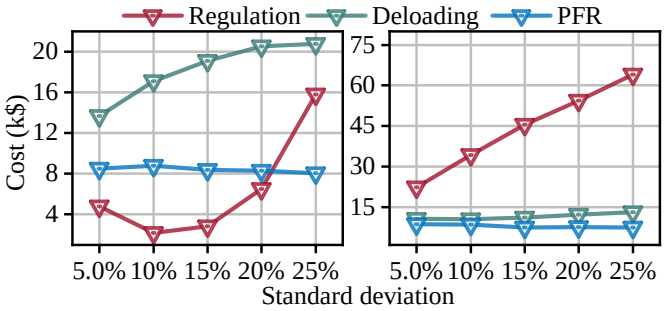

Fig. 11. Comparison of regulation reserve, wind deloading and PFR reserve cost with (left) and without (right) WTs providing upward regulation reserves under different standard deviation settings.

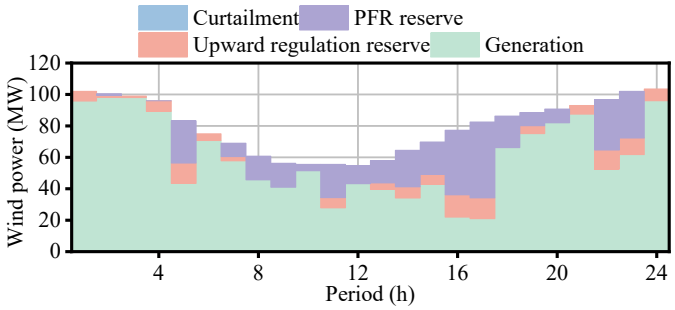

Fig. 12. Power allocation of WT 2.

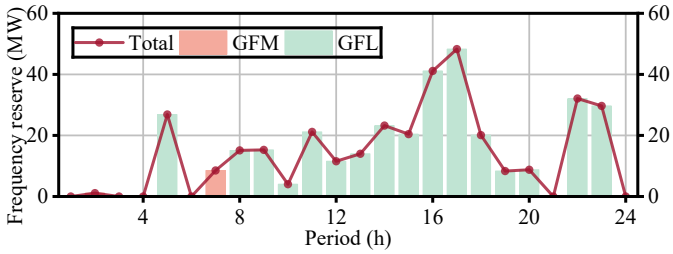

Fig. 13. Frequency reserves of WT 2 under GFM/GFL modes.

sample standard deviation. Fig. 10 presents the comparison of total costs. It can be observed that for all standard deviation settings, the total cost with WTs providing upward regulation reserves is lower than that without. Moreover, as the standard deviation increases, indicating higher wind power uncertainty, the cost increase in the case without WTs providing upward regulation reserves becomes more significant. This implies that fully utilizing the flexibility of WTs effectively reduces operational costs, particularly in high-uncertainty scenarios.

Furthermore, we illustrate the trends of regulation reserves, PFR reserves, and wind deloading costs, as shown in Fig. 11. It can be seen that the cost trends of PFR reserves remain similar in both cases, whereas the trends of the other two cost components differ significantly. First, considering the proposed method (left figure), the cost of regulation reserves initially decreases and then increases with increasing standard deviation, while deloading costs continuously increase. This is because, as wind power uncertainty intensifies, the system requires more regulation reserves to manage wind power UR, and WTs need to deload to reduce their UR response. As a combined effect, WTs' deloading increases, while the portion of deloading allocated to upward regulation reserves is used to mitigate the UR of other WTs, reducing the demand for CGU regulation reserves. Consequently, the cost of regulation reserves initially declines. However, as wind power uncertainty further increases, the upward regulation reserves provided by WTs become unreliable, necessitating CGUs to maintain system regulation capability, leading to an increase in regulation reserve costs.

In contrast, when WTs are not considered for providing upward regulation reserves, the system's regulation reserve cost exhibits an approximately linear increase as wind power uncertainty intensifies. Meanwhile, deloading costs also rise as WTs reduce their UR response to alleviate the burden on CGUs.

Furthermore, we present the scheduling results of an individual WT. Taking WT 2 connected at Bus 4 as an example, Fig. 12 illustrates its power partitioning results over the 24 scheduling periods. It can be observed that the WT output is explicitly decomposed into PFR reserve and upward reserve to support other WTs, while the curtailment component intended to mitigate its own uncertainty is not activated.

Furthermore, to illustrate the flexibility of the control-mode modeling, Fig. 13 presents the frequency reserve scheduling results of WT 2 under different control modes over the 24 periods. It can be seen that the GFM and GFL modes are adaptively selected across different periods and are mutually exclusive, which verifies the correctness and effectiveness of the proposed modeling formulation.

F. Comparison with Other FCS Methods

1) Comparison with the Direct DNN-Embedding Method: In this paper, the DNN-based FN constraint is incorporated through an iterative procedure using the gradients of the

trained DNN model. To verify the effectiveness of the proposed approach, a comparison is conducted with the method that directly embeds the DNN into the scheduling model. Table IV reports the total operating cost, the computational time, as well as the number of variables and constraints involved in the scheduling model under the two FCS methods.

As can be observed, compared with the direct DNN-embedding method, the proposed iterative approach achieves an almost identical total cost, with a relative error of only 0.07%. Meanwhile, the computational time is reduced by approximately 237 times. This significant acceleration mainly stems from the fact that directly embedding the DNN-based FN constraints introduces a large number of binary variables and big-M constraints to linearize the neural network structure. In contrast, by iteratively incorporating the FN constraints through DNN gradients, the proposed method avoids explicitly embedding the entire DNN into the scheduling model.

TABLE IV
COMPARISON BETWEEN THE DIRECT DNN-EMBEDDING METHOD AND THE PROPOSED METHOD (M2)

Item	Direct DNN embedding	Proposed method
Total cost (k$)	294.55	294.76
Computational time (s)	3786.5	15.96
Number of variables	39072	16008
Number of constraints	31109	19673

2) Comparison with Analytical-Based FN Methods: Furthermore, the proposed method is compared with the methods in [4], [5]. Table V summarizes the computational results of these methods, including the total operating cost, computational time, and the average FN over the 24 scheduling periods. As can be observed, the proposed method yields the lowest total cost, mainly because a more accurate representation of the FN constraint is adopted. In comparison, both [4] and [5] rely on simplified modeling assumptions. Specifically, the method in [4] fails to capture the heterogeneous PFR response characteristics of different resources, leading to an underestimation of the frequency support capability of WTs. The method in [5], although accounting for resource heterogeneity, neglects system damping and the droop-based frequency response provided by WTs, which also introduces conservativeness. By learning the complete frequency response behavior, the proposed method achieves FN values that remain above 49.5 Hz across all 24 periods without excessive margins, thereby ensuring frequency security while significantly improving economic performance.

TABLE V
COMPARISON WITH ANALYTICAL-BASED FN METHODS

Item	Method [4]	Method [5]	Proposed method
Total cost (k$)	328.92	318.87	294.76
Computational time (s)	2.95	67.62	15.96
Average FN (Hz)	49.8721	49.6547	49.5027

G. Validation of the DRCC Method

1) Sensitivity Analysis with Respect to the Wasserstein Radius: A sensitivity analysis is conducted to verify the rationality of selecting $\theta = 0.5$ in this case study. Specifically, by varying θ from 0.3 to 0.7, we evaluate the total operating cost of case M2 and the out-of-sample violation probability of the chance constraints, which is quantified by the maximum empirical violation probability (MEVP). The MEVP is computed as [33]:

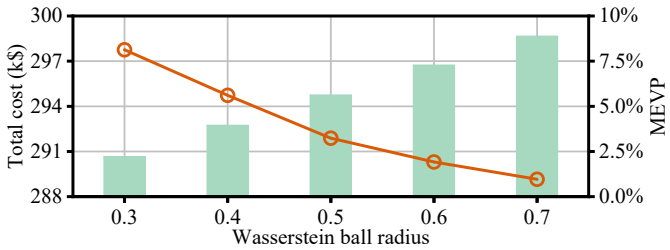

Fig. 14. Total cost and MEVP of M2 under different Wasserstein radius settings.

$$\mathrm{MEVP} = \max_k \left\{ \frac{1}{N_s} \sum_{s=1}^{N_s} \mathbb{I}\Big(a_k^{\mathrm{T}}(x)\, \xi^{(s)} + b_k(x) > 0 \Big) \right\},$$

where $\{\xi^{(s)}\}_{s=1}^{N_s}$ denotes the out-of-sample uncertainty realizations, $\mathbb{I}(\cdot)$ is the indicator function, and k indexes the chance constraints.

Fig. 14 illustrates the total cost and MEVP of M2 under different Wasserstein radius settings. It can be observed that, as the Wasserstein radius increases, the total operating cost rises while the MEVP gradually decreases. This is because a larger radius encompasses a wider set of potential probability distributions, thereby enhancing the robustness of the decision and reducing the out-of-sample violation probability of the chance constraints. When $\theta = 0.3$ and $\theta = 0.4$, an optimistic estimation of the uncertainty distribution leads to MEVP values exceeding the 5.0% threshold. In contrast, when $\theta = 0.5$, the MEVP is reduced to 3.23%. Further increasing the Wasserstein radius would result in higher operating costs with only marginal improvements in robustness. Therefore, considering both economic efficiency and robustness, $\theta = 0.5$ is selected as a compromise in this study.

Furthermore, we investigate the variations in regulation reserve cost and WTs deloading cost under different Wasserstein radius settings, as illustrated in Fig. 15. Interestingly, contrary to intuition, the regulation reserve cost does not increase monotonically with the Wasserstein radius and even decreases when $\theta = 0.7$. Similarly, the WTs deloading cost is also non-monotonic and exhibits a reduction at $\theta = 0.6$. This behavior indicates that, under different risk preference levels, the system dynamically balances uncertainty mitigation by reallocating between regulation reserve provision and WTs deloading. Such results further highlight the importance of explicitly modeling the flexibility of WTs. Although individual cost components vary non-monotonically with the Wasserstein radius, the overall total cost and MEVP exhibit monotonic trends.

2) Comparison with Other Chance-Constrained Programming Methods: To demonstrate the advantages of the Wasserstein-distance-based DRCC method adopted in this

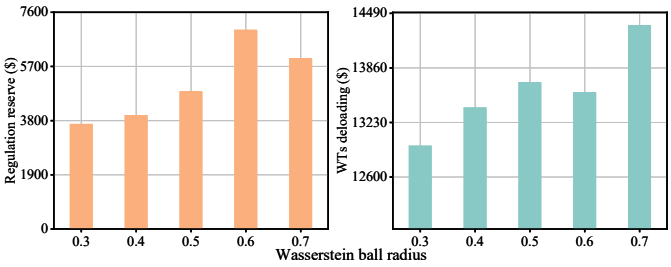

Fig. 15. Regulation reserves and WTs deloading cost of M2 under different Wasserstein radius settings.

paper, we compare it with two commonly used chance-constrained programming (CCP) approaches, namely the Gaussian-based CCP (G-CCP) [24], which relies on a Gaussian distribution assumption, and the moment-based DRCC method (M-DRCC) [38].

Table VI reports the performance of G-CCP, M-DRCC, and the proposed DRCC method for M2. The M-DRCC method achieves zero empirical violations but results in the highest total cost, mainly because the moment-based ambiguity set relies only on first- and second-order statistics, leading to an overly conservative uncertainty representation and excessive reserve allocation. In contrast, the G-CCP method depends on a Gaussian distribution assumption, which tends to be overly optimistic and may cause poor out-of-sample performance when the true distribution deviates from the assumed form. Compared with these methods, the Wasserstein-distance-based DRCC approach provides a better balance between robustness and economic efficiency by explicitly accounting for distributional deviations around empirical data, thereby reducing violation risk without introducing excessive conservativeness.

TABLE VI
COMPARISON WITH DIFFERENT CHANCE-CONSTRAINED PROGRAMMING METHODS

Method	M-DRCC	G-CCP	Proposed method
Total cost (k$)	340.30	282.72	297.76
MEVP (%)	0.00	22.88	3.23

3) MEVP under Different Out-of-Sample Testing Distributions: In the previous analysis, the MEVP values were evaluated using data-driven out-of-sample samples generated from a Gaussian distribution. To further examine whether the MEVP remains below the 5% threshold when the testing samples are not Gaussian-distributed, thereby validating the robustness of the proposed DRCC method, additional out-of-sample datasets are constructed. Specifically, 1980 samples are generated from Weibull, Beta, and Gamma distributions, respectively. In addition, real wind power forecast error data from three wind farms in Guangdong Province, China, are employed.

Table VII reports the MEVP results under different out-of-sample testing distributions. It can be observed that the MEVP remains below the prescribed 5% threshold across all testing scenarios, including non-Gaussian distributions and real wind power forecast error data. This demonstrates that the proposed DRCC method maintains a stable level of robustness even when the out-of-sample uncertainty deviates significantly from the assumed distribution, thereby confirming its effectiveness and practical applicability.

TABLE VII
MEVP UNDER DIFFERENT OUT-OF-SAMPLE TESTING DISTRIBUTIONS

Sample type	Gaussian	Weibull	Beta	Gamma	Real data
MEVP (%)	3.23	4.85	3.99	3.67	4.32

VI. CASE STUDY II: MODIFIED IEEE 118-BUS TEST SYSTEM

To further verify the scalability of the proposed UFCS-DNN method, case studies are conducted on the modified IEEE 118-bus test system. This system consists of 54 CGUs and 5 WTs, with installed capacities of 10,370 MW and 3,600 MW, respectively. The load demand ranges from 4,000 MW to 8,000 MW.

WTs are assumed to have deadband and delivery times of 0.1 s and 2 s, respectively, while CGUs are categorized into nine groups with deadband times of 0.5, 0.6, 0.7, 0.8, 0.9, 1, 1.2, 1.4, and 1.6 s, and delivery times of 2, 5, 6, 7, 8, 9, 10, 12, 14, and 16 s. Consequently, a total of 10 types of PFR products are considered. The LHS sampling ranges of the corresponding variables are summarized in Table VIII. The feature dimension for DNN training is set to 12, and all other parameters remain consistent with the modified IEEE 5-bus test system.

Table IX presents the economic costs and CPU time for M1 and M2 in the modified IEEE 118-bus test system. It can be observed that the total cost of M2 increases by 17.08% compared to M1, with the deloading cost experiencing the most significant rise, reaching 2.93 times that of M1. Unlike the results in the modified IEEE 5-bus test system, the PFR reserve cost of CGUs in M2 is 1.33% higher than in M1. This increase is attributed to the consideration of multiple CGU delivery capabilities, leading to the deployment of faster-responding but more expensive CGUs for PFR reserves, as illustrated in Fig. 16. Additionally, M2 commits more CGUs than M1, resulting in higher no-load and start-up/shutdown costs. Moreover, constrained by minimum power limits and increased WTs' power curtailment, the generation cost of CGUs also increases.

In terms of CPU time, M2 requires 825 seconds more than M1, which is acceptable for the day-ahead scheduling problem. Moreover, numerous recently developed acceleration algorithms for mixed-integer linear programming can be integrated with the proposed method to further enhance computational efficiency.

The FN across time periods under M1 and M2 is shown in Fig. 17. It can be observed that the proposed method strictly maintains FN above 49.5 Hz without exceeding it excessively, ensuring non-conservativeness. In contrast, without considering FN constraints, frequency security cannot be guaranteed.

We further demonstrate the advantages of the proposed acceleration algorithm in a larger system, as shown in Fig. 18.

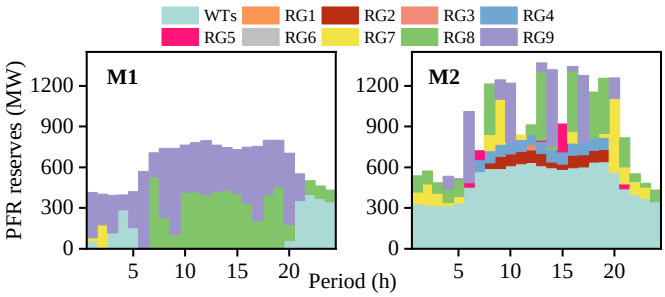

Fig. 16. PFR reserves deployment of M1 and M2 in the modified IEEE 118-bus test system, where RG1 to RG9 represent nine types of CGU PFR services arranged in ascending order of delivery time.

It can be observed that the proposed method maintains both efficiency and economic performance. In contrast, excluding the biased SVM results in suboptimal economic benefits, requires more iterations, and exhibits oscillations during the process.

We present the variation of FN across time periods during the iterative process, as shown in Fig. 19. It can be observed that the early morning and late-night periods (0–8h and 20–24h) become the key periods restricting the convergence speed. This is because these periods have a high wind power penetration, leading to insufficient system inertia, severe frequency security risks, and strong nonlinearity in the frequency security boundary, necessitating multiple iterations. Additionally, alternating occurrences of satisfied and unsatisfied FN constraints are observed, which can be attributed to the existence of inter-period coupling constraints and the inherent inaccuracies of machine learning methods.

TABLE VIII
SAMPLING RANGES OF DNN INPUT FEATURES IN THE MODIFIED IEEE 118-BUS SYSTEM

Variable	Symbol	Range
System inertia	H^{sys}	5–15 s
System damping	D^{sys}	0.03–0.20 p.u.
WT PFR reserve	R^W	0–0.07 p.u.
CGU PFR reserve (type 1)	R_1^G	0–0.07 p.u.
CGU PFR reserve (types 2–9)	R_2^G–R_9^G	0–0.05 p.u.

TABLE IX
COST COMPARISON OF DIFFERENT METHODS IN THE MODIFIED IEEE 118-BUS TEST SYSTEM

Cost (k$)	M1	M2
No load	120.56	141.15
Start-up/Shut-down	10.23	13.70
Generation	1812.23	2031.10
CGUs Regulation reserve	50.05	57.94
CGUs PFR reserve	172.28	174.57
WTs deloading	109.27	320.34
Total	2274.63	2738.80
CPU time (s)	325.28	1149.57

VII. CONCLUSIONS

This paper proposes a UFCS-DNN method, which iteratively incorporates linear surrogates guided by DNN gradients

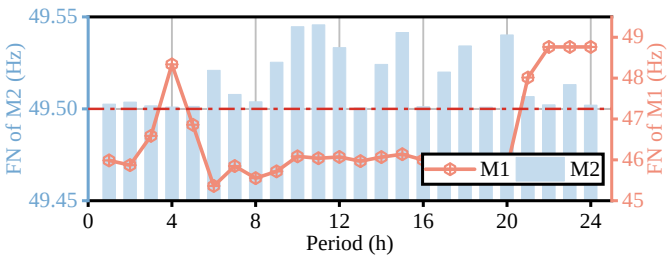

Fig. 17. FN under preset power imbalance for M1 and M2 in the modified IEEE 118-bus test system.

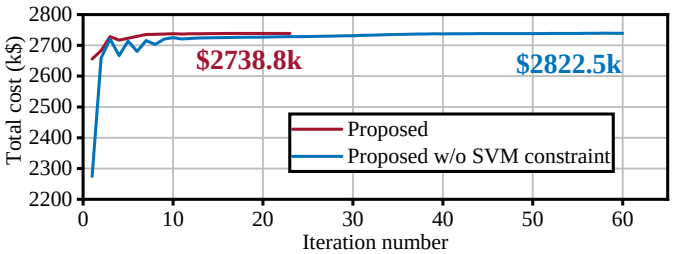

Fig. 18. Iterative process under different iterative methods.

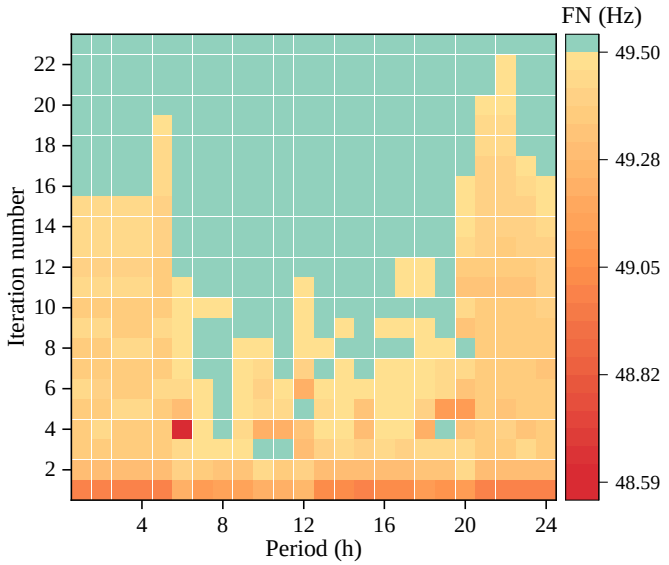

Fig. 19. FN across time periods during the iterative process in the modified IEEE 118-Bus test system.

to progressively satisfy the FN constraint. The deloaded power of WTs is utilized to respond to uncertainties and provide PFR reserves. Numerical results on the modified IEEE 5-bus and IEEE 118-bus test systems lead to the following conclusions:

1) Considering the FN constraint is crucial for low-inertia power systems. Compared with existing methods, the proposed approach guarantees that the FN in all 24 scheduling periods remains no lower than 49.5 Hz under the prescribed disturbances, at the expense of cost increases of 7.50% and 17.08% in the modified IEEE 5-bus and IEEE 118-bus systems, respectively.
2) The DNN-gradient-based linear FN surrogate preserves the high approximation accuracy of the DNN while sub-

stantially reducing the computational burden associated with direct DNN embedding. Results on the IEEE 5-bus test system demonstrate a 237-fold improvement in computational efficiency compared with MIP-style DNN embedding methods. In addition, the proposed biased-SVM initialization and tailored step-size strategy enable fast convergence, requiring only 15 and 23 iterations for the two test systems, respectively.

3) Exploiting the flexibility of WTs is of great importance. The results indicate that appropriately allowing WTs to curtail their output can effectively alleviate the burden of mitigating uncertainty. As the uncertainty level increases, the proposed method achieves increasing cost reductions compared with methods that do not consider WT flexibility, with total cost savings rising from 1.44% to 6.11%.

Appendix A
Derivation of the FN Constraint

The specific expressions for each parameter in (3) are as follows:

$$\begin{cases} \lambda_1 = e^{(b/a)t_1}\Big(\frac{\eta_1}{b}\big(1-e^{-(b/a)t_1}\big) - \frac{\nu_1}{b}t_1 - \frac{\eta_2 b - a\nu_1}{b^2}\Big), \\ \lambda_{i+1} = \lambda_i + e^{(b/a)t_{i+1}}\Big(\frac{\nu_i}{b}t_{i+1} + \frac{\eta_{i+1}b - a\nu_i}{b^2} \\ \qquad\qquad -\frac{\nu_{i+1}}{b}t_{i+1} - \frac{\eta_{i+2}b - a\nu_{i+1}}{b^2}\Big), \\ \lambda_{2N} = \lambda_{2N-1} + e^{(b/a)t_{2N}}\Big(\frac{\nu_{2N-1}}{b}t_{2N} \\ \qquad\qquad +\frac{\eta_{2N}b - a\nu_{2N-1}}{b^2} - \frac{\eta_{2N+1}}{b}\Big), \end{cases} \tag{30}$$

$$\begin{cases} a = \dfrac{2H^{sys}}{f_0}, \\ b = D^{sys}, \end{cases} \tag{31}$$

$$\begin{cases} \nu_i = \dfrac{R_1}{T_1}, \qquad i \in [1,\, N], \\ \nu_{N+i} = \displaystyle\sum_{k=1}^{N}\frac{R_k}{T_k} - \sum_{k=1}^{i}\frac{R_k}{T_k}, \qquad i \in [1,\, N-1], \end{cases} \tag{32}$$

$$\begin{cases} \eta_1 = -\Delta P^{imb}, \\ \eta_{i+1} = -\displaystyle\sum_{k=1}^{i}\tfrac{t_k}{T_k}R_k - \Delta P^{imb}, \qquad i \in [1,\, N], \\ \eta_{i+N+1} = -\displaystyle\sum_{k=1}^{N}\tfrac{t_k}{T_k}R_k - \Delta P^{imb} \\ \qquad\qquad + \displaystyle\sum_{k=1}^{i}\Big(\tfrac{t_k}{T_k}R_k + R_k\Big), \qquad i \in [1,\, N] \end{cases} \tag{33}$$

$$\begin{cases} \tau_i = t_i, \qquad i \in [1,\, N], \\ \tau_{N+i} = t_i + T_i, \qquad i \in [1,\, N], \end{cases} \tag{34}$$

where t_i and T_i are the dead band and delivery time of PFR service i.

As the expression of frequency deviation is piecewise-defined, its derivative cannot be obtained in a closed form. Therefore, the FN constraint can be reformulated by requiring that, within a given response interval $[0,\, \tau_{\mathrm{PFR}}]$ (typically $\tau_{\mathrm{PFR}} = 30\,\mathrm{s}$), the absolute value of the frequency deviation does not exceed the threshold $\Delta f_{\max}$. That is, the FN constraint can be equivalently written as

$$-\Delta f_{\max} \;\le\; \Delta f(\tau) \;\le\; \Delta f_{\max}, \qquad \forall \tau \in [0,\, \tau_{\mathrm{PFR}}]. \tag{35}$$

By examining the specific expressions of all parameters in (3), we observe that, for each scheduling period t and under a given disturbance, the extremum of $\Delta f(\tau)$ over the interval $[0,\, \tau_{\mathrm{PFR}}]$ (i.e., the FN) depends on the decision variables H_t^{sys}, D_t^{sys}, $\mathbf{R}_t^G$, and $\mathbf{R}_t^W$. Therefore, the FN constraint can be written in a compact form as (9).

Appendix B
Reformulation of Constraints (20)

The deterministic part of (20) is given as follows:

$$\sum_{g\in G} P_{g,t} + \sum_{w\in W} W_{w,t} = \sum_{d\in D} P_{d,t}^{D}, \forall t \tag{36}$$

The stochastic term in (20) is given as follows, where we omit the notation "$\forall t$ " for brevity:

$$\mathbf{1}_{|\mathcal{G}|\times 1}^{\mathrm{T}}\mathbf{A}_t\varpi_t + \mathbf{1}_{|\mathcal{W}|\times 1}^{\mathrm{T}}\mathbf{B}_t\varpi_t = \mathbf{1}_{|\mathcal{W}|\times 1}^{\mathrm{T}}C_t\varpi_t \tag{37}$$

where :

$$\begin{aligned} \mathbf{A}_t\varpi_t &= \begin{pmatrix} \alpha_{1,1,t} & \cdots & \alpha_{1,|\mathcal{W}|,t} \\ \vdots & \cdots & \vdots \\ \alpha_{|\mathcal{G}|,1,t} & \cdots & \alpha_{|\mathcal{G}|,|\mathcal{W}|,t} \end{pmatrix} \begin{bmatrix} \varpi_{1,t} \\ \vdots \\ \varpi_{|\mathcal{W}|,t} \end{bmatrix} \\ &= \begin{pmatrix} \alpha_{1,t}^{\mathrm{T}}\varpi_t \\ \vdots \\ \alpha_{|\mathcal{G}|,t}^{\mathrm{T}}\varpi_t \end{pmatrix} \end{aligned} \tag{38}$$

$$\begin{aligned} \mathbf{B}_t\varpi_t &= \begin{pmatrix} 0 & \cdots & \beta_{1,|\mathcal{W}|,t} \\ \vdots & \mathbf{0} & \vdots \\ \beta_{|\mathcal{W}|,1,t} & \cdots & 0 \end{pmatrix} \begin{bmatrix} \varpi_{1,t} \\ \vdots \\ \varpi_{|\mathcal{W}|,t} \end{bmatrix} \\ &= \begin{pmatrix} \beta_{1,t}^{\mathrm{T}}\varpi_t \\ \vdots \\ \beta_{|\mathcal{W}|,t}^{\mathrm{T}}\varpi_t \end{pmatrix} \end{aligned} \tag{39}$$

$$\begin{aligned} C_t\varpi_t &= \begin{pmatrix} 1-\delta_{1,t} & \mathbf{0} & 0 \\ \mathbf{0} & \ddots & \mathbf{0} \\ 0 & \mathbf{0} & 1-\delta_{|\mathcal{W}|,t} \end{pmatrix} \begin{bmatrix} \varpi_{1,t} \\ \vdots \\ \varpi_{|\mathcal{W}|,t} \end{bmatrix} \\ &= \begin{pmatrix} (1-\delta_{1,t})\varpi_{1,t} \\ \vdots \\ (1-\delta_{|\mathcal{W}|,t})\varpi_{|\mathcal{W}|,t} \end{pmatrix} \end{aligned} \tag{40}$$

By observation, we find that a sufficient condition for satisfying (37) is that each row of $\mathbf{A}_t + \mathbf{B}_t - C_t$ equals zero. Thus, we have:

$$\sum_{g\in\mathcal{G}} \alpha_{g,w,t} + \sum_{j\in\mathcal{W}} \beta_{j,w,t} = 1 - \delta_{w,t}, \forall w, t \tag{41}$$

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